INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 1

CANDIDATE NAME

CLASS | INDEX NUMBER

CHEMISTRY

8872/02

Paper 2

11 September 2012

Section A: Structured Questions
Section B: Free Response Questions

2 hours

Candidates answer Section A on the Question Paper
Additional Materials: Answer Paper
Data Booklet
Graph Paper

READ THESE INSTRUCTIONS FIRST

Write your index number, name and civics group.
Write in dark blue or black pen.
You may use pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A: Structured Questions (40m)
Answer **all** questions in the space provided.

Section B: Free Response Questions (40m)
Answer **two** questions on separate writing papers.

You are advised to show all working in calculations.
You are reminded of the need for good English and clear presentation in your answers.
You are reminded of the need for good handwriting.
Your final answers should be in 3 significant figures.

You may use a calculator.

The number of marks is given in brackets [ ] at the end of each question or part question.
At the end of the examination, fasten all your work securely together.

For Examiner's Use	
Section A	
1	11
2	9
3	7
4	13
Significant figures	
Handwriting	
Total	40

This document consists of **13** printed pages and **1** blank page.

For Examiner's Use

Section A

Answer ALL questions on the spaces provided.

1. **(a)** The elements in Period 3 burns with fluorine to produce the corresponding fluorides.

The formulae and melting points of the fluorides of the elements in Period 3, Na to C*l* are given in the table.

Formula of fluoride	NaF	MgF_2	AlF_3	SiF_4	PF_5	SF_6	ClF_5
m.p. / K	1268	990	1017	183	139	223	170

(i) Describe the bonding in the compound sodium fluoride. Draw a diagram to illustrate your answer.

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

……………………………………………………………………………………[2]

(ii) Complete the electronic configuration for the following species.

the A*l* atom $1s^2$……………………………………………………

the F^- ion $1s^2$……………………………………………………

[2]

(b) The elements burnt in oxygen to produce oxides.

(i) Describe what would be observed when sulfur reacts with oxygen.

…………………………………………………………………………………………

……………………………………………………………………………………[1]

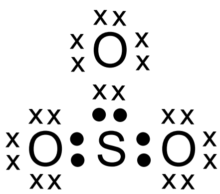
(ii) Write a balanced equation, including state symbols, for the reaction in **(i)**.

……………………………………………………………………………………[1]

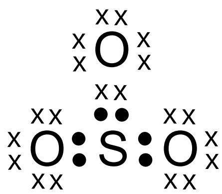
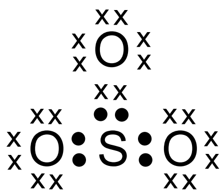
For Examiner's Use

(iii) The oxide formed in **(ii)** can be further oxidised to SO_3 by adding an appropriate catalyst, V_2O_5.

A 'dot-and-cross' diagram of a SO_3 molecule is shown below. Only electrons from outer shells are represented.



In the table below, there are two copies of this structure.
On the structures, draw a circle around a pair of electrons that is associated with **each** of the following.

(I) A co-ordinate bond	(II) a covalent bond
	

[1]

(c) An unstable isotope phosphorus-35, ^{35}P, produces sulfur-35, ^{35}S when a neutron is converted to a proton. This process is known as beta emission.

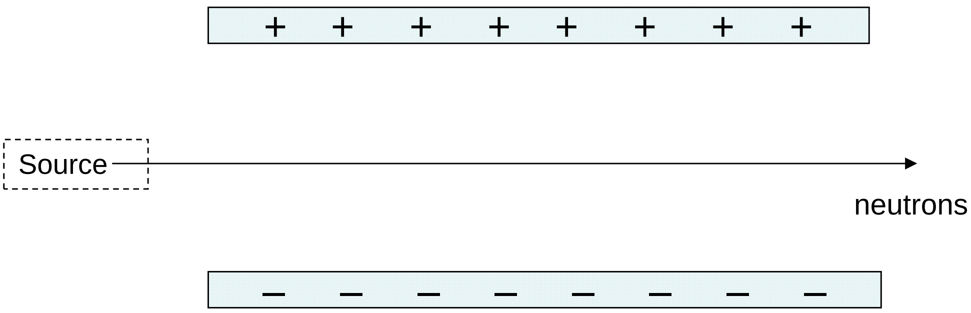
(i) Deduce the number of protons and neutrons isotope sulfur-35, ^{35}S contains.

protons ………………………..

neutrons ………………………. [2]

(ii) Sketch on the diagram below how beams of each of the following particles are affected by the electric field, labelling each beam clearly.

I. electrons
II. protons

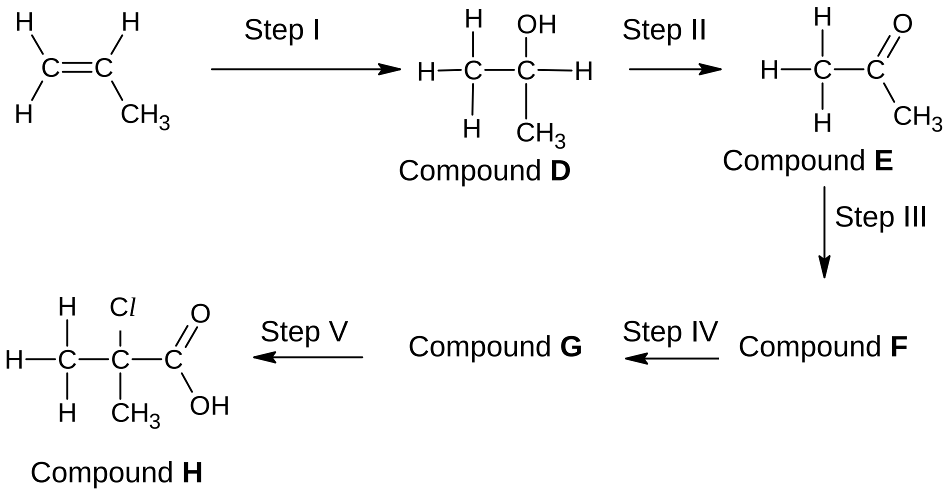


[2]

[Total:11]

For Examiner's Use

2 Prop-1-ene is an important raw material for the production of polypropylene, a recyclable polymer commonly used to make plastic containers. Compound **I** can be produced from prop-1-ene in the following synthesis pathway.



(a) Suggest reagents and conditions for steps I and II.

Step I

Reagent and Condition: ………………………………………………………………

Step II

Reagent and Condition: ………………………………………………………………

[2]

(b) Propose a three-step synthesis for the conversion of Compound **E** to Compound **H**, giving the structural formulae of the intermediates, Compound **F** and Compound **G** in the space provided.

Step III

Reagent and Condition: ..……………………………………………………………

Step IV

Reagent and Condition:...……………………………………………………………

Step V

Reagent and Condition:...…………………………………………………………..

Compound **F**	Compound **G**

[5]

For Examiner's Use

(c) Suggest a simple chemical test to distinguish Compound **D** and propanol. State what would be observed during the test.

Reagent and Condition: ..……………………………………………………………

Observation for

Compound **D**: ……………………………………………………………………….

Propanol:……………………………………………..…………………………….

[2]

[Total: 9]

3 **(a)** In theory, the reaction shown below could be used to prepare SiF_4.

$$CF_4(g) + SiH_4(g) \longrightarrow CH_4(g) + SiF_4(g)$$

The table below shows the bond enthalpy data for covalent bond in its gaseous phase.

Gas–phase bond	Average bond enthalpy / $kJ\ mol^{-1}$
C–H	413
C–F	467
Si–H	318
Si–F	553

Calculate the enthalpy change of the reaction, including the units, shown in the given equation.

[2]

For Examiner's Use

(b) The reaction shown in **(a)** is not observed to take place at room temperature and pressure. However, recent research has revealed a method of exchanging a fluorine atom bonded to carbon with a hydrogen atom bonded to silicon at room temperature and pressure. This was accomplished by introducing triethylsilyl cations, $(C_2H_5)_3Si^+$. The proposed mechanism contains the following steps.

Step 1 $(C_2H_5)_3Si^+ + C_6H_5CF_3 \longrightarrow (C_2H_5)_3SiF + C_6H_5CF_2^+$
Step 2 $(C_2H_5)_3SiH + C_6H_5CF_2^+ \longrightarrow (C_2H_5)_3Si^+ + C_6H_5CF_2H$

(i) Write down the overall equation shown by the reaction steps above.

……………………………………………………………………………….. [1]

(ii) State and explain the role of $(C_2H_5)_3Si^+$ in this process.

…………………………………………………………………………………

…………………………………………………………………………………

……………………………………………………………………………….. [2]

(c) Suggest why Si–F bonds have a higher average bond enthalpy than C–F bonds.

…………………………………………………………………………………………

……………………………………………………………………………………….. [1]

(d) Suggest why, despite the stronger Si–F bond, SiF_4 is more reactive than CF_4.

…………………………………………………………………………………………

……………………………………………………………………………………….. [1]

[Total: 7]

For Examiner's Use

4 **(a)** Carbon monoxide, which can be used to make methanol, may be formed by reacting carbon dioxide with hydrogen.

$$CO_2(g) + H_2(g) \rightleftharpoons CO(g) + H_2O(g) \qquad K_c = 1.44 \text{ at } 1200 \text{ K}$$

(i) Write K_c expression for the above reaction, stating its units.

[1]

(ii) A mixture containing 0.50 mol of CO_2, 0.50 mol of H_2, 0.20 mol of CO and 0.20 mol of H_2O was placed in a 2.0 dm^3 flask and allowed to come to equilibrium at 1200 K.

Calculate the amount, in moles, of each substance present in the equilibrium mixture at 1200 K.

[5]

For Examiner's Use

(b) **(i)** Write an equation for the dissociation of water.

……………………………………………………………………………………. [1]

(ii) Use the equation in **(i)** to write an expression for the equilibrium constant, K_c, for this reaction.

[1]

(iii) Use the expression in **(ii)** to show that $K_w = [H^+][OH^-]$. Justify and explain your reasoning.

…………………………………………………………………………………………

…………………………………………………………………………………………

……………………………………………………………………………………. [2]

(iv) At 373K, the ionic product of water, K_w, has a value of 51.3×10^{-14} mol^2 dm^{-6}. Use this information to calculate the pH of water at 373 K.

[1]

(v) At 298K, the pH of pure water is measured to be 7.0. When pure water is warmed to 338K, it is observed that its pH drops to 6.7.
Find the pK_w of pure water at 338 K.

[1]

(vi) Using the information in **(v)**, calculate the pH of 0.100 mol dm^{-3} aqueous sodium hydroxide at 338K.

[1]

[Total: 13]

Section B: Free Response Questions

Answer **two** of the three questions in this section on separate writing papers.
Begin each answer on a fresh sheet of paper.

5 **(a)** The concentration of a hydrogen peroxide solution can be determined by titration with acidified potassium manganate(VII) solution. In this reaction the hydrogen peroxide is oxidised to oxygen gas.

A 5.00 cm^3 sample of the hydrogen peroxide solution was added to a volumetric flask and made up to 250 cm^3 of aqueous solution. A 25.0 cm^3 sample of this diluted solution was acidified and reacted completely with 24.35 cm^3 of 0.0187 $mol dm^{-3}$ potassium manganate(VII) solution.

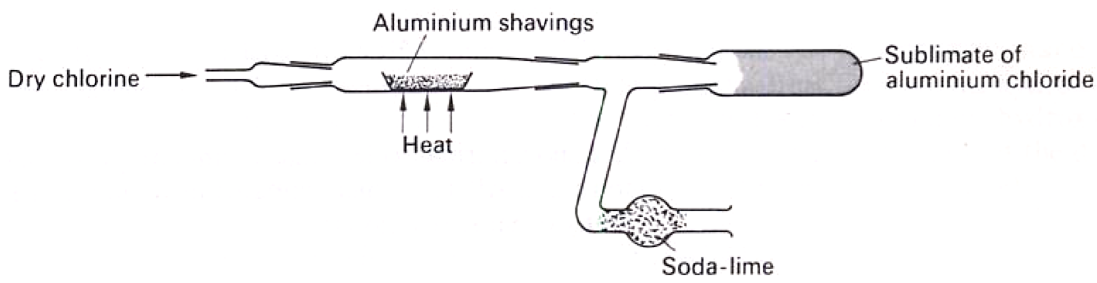
(i) Write an equation for the reaction between acidified potassium manganate(VII) solution and hydrogen peroxide.

(ii) Use this equation and the results given to calculate a value for the concentration, in $mol dm^{-3}$, of the original hydrogen peroxide solution.

(If you have been unable to write an equation for this reaction you may assume that 3 mol of $KMnO_4$ react with 7 mol of H_2O_2. This is **not** the correct reacting ratio.)

[5]

(b) When dry chlorine is passed over heated aluminium foil in a hard glass tube, a vapour is produced which condenses to a yellow-white solid in the cooler parts of the tube. At low temperatures the vapour has the empirical formula $AlCl_3$ and M_r 267.



(i) Suggest the molecular formula of the vapour, and draw a structure to describe its bonding.

The yellow-white solid reacts with water in two separate ways.

- When a few drops of water are added to the solid, steamy fumes are evolved and a white solid remains, which is insoluble in water.

- When a large amount of water is added to the solid, a clear, weakly acidic solution results.

(ii) Write the equations, including state symbols, for these two reactions.

[6]

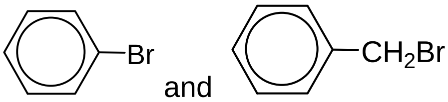
(c) The halogenoethanes C_2H_5Cl, C_2H_5Br and C_2H_5I differ in their physical properties and reactivities.

(i) Suggest and explain how the boiling points of these compounds differ.

(ii) Describe and explain how the reactivities of the three compounds differ towards nucleophilic reagents.

[3]

(d) Suggest simple test-tube reactions by which the isomers in the following pair can be distinguished from each other. You should state the reagents and conditions for the test, and describe how each of the isomers in the pair behaves.



[2]

(e) Chlorofluoroalkanes, CFCs, were once used as refrigerant fluids and aerosol propellants. In many applications they now have been replaced by alkanes. This is because CFCs contribute to the destruction of the ozone layer.
Ultraviolet radiation from the sun in the stratosphere photolyses CFC molecules to form chlorine radicals, which initiate a chain reaction of destruction of ozone molecules.

(i) Suggest one reason why CFCs were originally used for these purposes.

(ii) Suggest **one** potential hazard of using alkanes instead of CFCs.

[2]

(f) One method of making 1-bromobutane in the laboratory is described below.

Stage 1 — Place 35 g of powdered sodium bromide, 30 cm^3 of water and 25 cm^3 (20 g) of butan-1-ol in a 250 cm^3 two necked flask fitted with a tap funnel and reflux condenser.

Stage 2 — Concentrated sulfuric acid (25 cm^3) is then placed in the tap funnel and added drop by drop to the reagents in the flask, keeping the contents well shaken and cooled occasionally in an ice-water bath.

The overall reaction may be considered to take place in two stages. In the first stage, sodium bromide and concentrated sulfuric acid react together to form HBr and a sodium salt. In the second stage, butan-1-ol reacts with the HBr that is formed in the first stage.

Write an equation for **each** of these stages.

[2]

[Total:20]

6 **(a)** The elements in Period 3 range from metals on the left of the Periodic Table to non-metals on the right.

Using **one** property of the element and **one** of its chlorides, describe **two** pieces of evidence which support each of the following statements.

(i) Sodium is a metal

(ii) Phosphorus is a non-metal

[4]

(b) State the shape and bond angle of the PF_3 molecule. Explain your answer using the Valence Shell Electron Pair Repulsion Theory.

[3]

(c) A neutral liquid **A**, $C_{10}H_{12}O_2$ when heated with aqueous hydrochloric acid gave two products, **B** and **C**.

B, $C_8H_{10}O$, gave a yellow precipitate when warmed with aqueous alkaline iodine. When heated with acidified sodium dichromate(VI), **B** gave **D**, C_8H_8O.

D reacts with 2,4-dinitrophenylhydrazine to give orange crystals, **E**.

Deduce the possible structures of compounds **A** to **E**. Explain the chemistry of the reactions involved. Write an equation for **A** when it is heated with aqueous hydrochloric acid.

[9]

(d) How would you expect the acidity of 3-chloropropanoic acid and of fluoroethanoic acid to compare with that of chloroethanoic acid?

[4]

[Total:20]

7 **(a)** The main buffering agent in blood plasma is the carbonic acid /hydrogen carbonate ion (H_2CO_3 / HCO_3^-) system. In blood plasma, the concentration of hydrogen carbonate ion is about twenty times the concentration of carbonic acid. The pH of arterial blood plasma is 7.40. If the pH falls below this normal value, a condition called *acidosis* is produced. If the pH rises above the normal value, the condition is called *alkalosis*.

Write equations to show how the carbonic acid /hydrogen carbonate ion buffer system works to maintain the pH of blood plasma close to 7.40 on the addition of

(i) H^+ ions
(ii) OH^- ions [2]

(b) Calculate the pH of a 0.30 mol dm^{-3} solution of sulfuric acid, H_2SO_4. [1]

(c) Propyl ethanoate undergoes a slow acid–catalysed hydrolysis in water.

$$CH_3COOCH_2CH_2CH_3 + H_2O \xrightarrow{HCl} CH_3COOH + CH_3CH_2CH_2OH$$

Two experiments were carried out, starting with different concentration of HC*l*. The following results were obtained.

time / minutes	Experiment 1, with [HC*l*] = 0.10 mol dm^{-3}	Experiment 2, with [HC*l*] = 0.05 mol dm^{-3}
	[$CH_3COOCH_2CH_2CH_3$] / moldm^{-3}	$CH_3COOCH_2CH_2CH_3$] / moldm^{-3}
0	0.0050	0.0050
15	0.0040	0.0045
30	0.0032	0.0040
45	0.0026	0.0036
60	0.0021	0.0032
75	0.0017	0.0029
90	0.0014	0.0026

(i) Using the same axes, plot graphs of [$CH_3COOCH_2CH_2CH_3$] against time for the two experiments.

(ii) Use your graphs to determine the order of reaction with respect to [$CH_3COOCH_2CH_2CH_3$] and to [HC*l*].

(iii) Construct a rate equation for the reaction and use it to calculate a value for the rate constant, giving its units.

[9]

(d) At room temperature, gaseous dinitrogen tetraoxide and nitrogen dioxide are in dynamic equilibrium according to the following equation:

$$N_2O_4\,(g) \rightleftharpoons 2NO_2\,(g) \qquad K_c = 7.246 \times 10^{-3}$$

Colourless brown

The mixture turns into a darker shade of brown when it was placed in hot water.

(i) Deduce and explain whether the forward reaction is endothermic or exothermic.

(ii) Explain, with reason, whether the dissociation of N_2O_4 is favoured by high or low pressure.

(iii) 1.38 g of N_2O_4 was placed in a 2.0 dm^3 reaction vessel. At equilibrium, it was found that N_2O_4 was 40% dissociated. Calculate the equilibrium concentration of NO_2.

(iv) Draw a labelled reaction pathway diagram for the dissociation of N_2O_4.

[8]

[Total:20]

END OF PAPER

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